

Advanced Mango Leaf Disease Detection and Severity Analysis with Federated Learning and CNN

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Abstract— Worldwide mango production is threatened by mango leaf diseases, resulting in considerable financial losses. Effective illness management and reduction depend on early identification and severity level categorization. This research provides a brand-new FL-CNN model for categorizing mango leaf diseases into four severity categories. The FL-CNN model allows training to be distributed across several clients without compromising data privacy. The suggested model consistently outperforms all clients in identifying the four severity levels of mango leaf diseases. The model's efficiency across customers with various data distributions is shown using macro, weighted, and micro-averaging techniques. Client 4 had the greatest macro average (96.11), weighted average (96.09), and micro average (96.08) values, while Client 1 had the lowest macro average (93.92), weighted average (93.96), and micro average (93.96) values. These findings show that the FL-CNN model successfully recognizes and categorizes various degrees of mango leaf disease severity, making it an invaluable tool for practical agricultural applications. Utilizing federated learning provides data privacy while offering a scalable and safe method for managing plant diseases. Additionally, federated learning preserves data privacy and lessens the need for data centralization, offering a solid foundation for cooperation between many stakeholders. Future studies should focus on enhancing the model's effectiveness and accuracy and investigating whether it can predict additional plant diseases and agricultural situations.

Keywords: *Mango leaf diseases, Convolutional neural networks (CNNs), Federated learning, Disease classification, Severity levels.*

I. INTRODUCTION

A tropical fruit tree with high economic and nutritional value, the mango (*Mangifera indica* L.) is extensively grown around the globe. However, several diseases, especially those that damage the leaves, may substantially impact the production and quality of mango harvests. These illnesses, which are brought on by pathogens like fungus, bacteria, and viruses, may significantly reduce crop yields and jeopardize the trees' health. To enact efficient control measures and maintain a sustainable agricultural system, timely diagnosis and precise categorization of mango leaf diseases and their severity levels are crucial. Four separate phases of mango leaf disease severity may be identified: (1) beginning, (2) mild, (3) moderate, and (4) severe. Each stage shows the disease's development and the severity of the leaf damage. It is essential to accurately classify these severity levels to choose the best management practices and lessen the effect on the mango crop. Convolutional Neural Networks (CNNs)

have been developed due to recent developments in computer vision and deep learning for various picture categorization applications, including plant disease detection. CNNs are a potential method for identifying illnesses in mango leaves because of their excellent accuracy and efficiency in detecting plant diseases. To train these models, a considerable quantity of data must be gathered and stored in a centralized repository, which might generate privacy and security issues, particularly when the data is sensitive or belongs to several parties. Federated learning is a distributed machine learning method that permits the localization of data while allowing the training of common models across several devices. CNN models [1] may be taught without needing a central repository for data storage by using federated learning, protecting data privacy and lowering data transmission costs. There are four key goals for this study. To further future research on mango leaf disease detection and classification, our first goal is to create a complete collection of annotated photos of mango leaves with disease kind and severity level. A unique Federated Learning-based Convolutional Neural Network (FL-CNN) [2] architecture will be created to categorize mango leaf diseases into four severity levels. Third, we will assess the suggested FL-CNN model's performance in contrast to conventional CNNs and other cutting-edge techniques, proving its potency in categorizing mango leaf illnesses and their severity levels. We will finally address possible difficulties, constraints, and future research prospects to further increase the precision and effectiveness of mango leaf disease classification using federated learning-based CNNs. This study suggests an FL-CNN—a convolutional neural network based on federated learning—to categorize mango leaf illnesses into four severity categories. The following are the primary contributions of this study paper:

Creation of a comprehensive collection of photos of mango leaves tagged with information about the disease's kind and severity level.

Design and implement a unique FL-CNN architecture emphasizing protecting data privacy and minimizing communication overhead, specially customized for classifying mango leaf disease.

A thorough analysis of the suggested FL-CNN model's performance compared to conventional CNNs and other cutting-edge techniques shows how well it can categorize mango leaf diseases and their severity levels.

To further increase the precision and effectiveness of mango leaf disease classification using federated learning-

based CNNs, it is possible to identify potential difficulties and limits of the proposed technique and provide ideas for future research paths.

A thorough description of the many ailments that harm mango leaves. One of the most prevalent diseases that damage mango trees is anthracnose, a fungus. *Colletotrichum gloeosporioides* brings it on and usually affects the tree's leaves, blooms, and fruit. Dark patches on the leaves, which may later become yellow or brown and cause leaf loss, are the initial symptoms of the illness. Black patches on infected flowers and fruit may acquire deformed and undesirable characteristics. Anthracnose, which spreads by wind and rain, is treatable and preventable with fungicides.

Powdery Mildew: This fungal disease causes the leaves to become deformed, stunted, and fall off the tree. It appears as a white, powdery material on the foliage. The fungus *Oidium Mangifera* causes it and may spread via wind or insects. Fungicides may be used to manage powdery mildew, which prefers warm, humid conditions.

Leaf Spot: This fungus, *Cercospora* sp., generate brown, circular spots on the leaves that, in extreme situations, may combine and result in defoliation. The disease may be managed using fungicides and is transmitted by water, wind, and insects.

Rust: This fungus infects the leaves, leaving orange or yellow patches that may turn them yellow and cause them to fall off the tree. The fungus *Puccinia Mangifera* is responsible for its cause and may spread via wind or insects. Rust can be managed using fungicides because it prefers warm, humid environments.

Mango Malformation: This viral illness results in twisted and deformed leaves, which may impede the development of the tree and diminish fruit output. The Mango Malformation Phytoplasma, spread by an insect vector, causes the illness. Mango malformation has no known treatment. Hence afflicted trees should be cut down to stop the illness from spreading to other trees. Black spots on the leaves are a symptom of the bacterial illness known as "bacterial black spots," which may also hasten the premature fall of the leaves. *Xanthomonas campestris* pv. *mangiferae* is the cause, and it may spread by wind or rain. Bacterial black spots are curable with copper-based fungicides and flourish in warm, humid environments.

II. LITERATURE REVIEW

The article describes how mango leaf diseases may reduce mango quality and productivity and how it might be challenging to diagnose them using conventional techniques effectively. Mango leaf illnesses have been classified by researchers using computer-aided and machine-learning techniques. However, these methods have certain drawbacks owing to increasing feature dimensionality, overfitting, computational complexity, and a lack of feature characteristics. The authors suggest a new mango leaf disease classification framework with four stages: data preparation, feature selection, learning and classification, and performance assessment. This framework aims to overcome these problems. To avoid overfitting and enhancing generalization, they chose 380 photos from the categories of healthy and ill mango leaves. They used a convolutional neural network with crossover-based levy flight distribution

to improve feature selection. For the learning stage, they used a pre-trained MobileNetV2 model. For the classification step, they used a support vector machine. Comparing the experimental findings to other cutting-edge techniques, they showed greater categorization performance [3]. This research compared the efficacy of fuzzy logic and convolutional neural networks (CNN) for diagnosing mango leaf illnesses. The researchers utilised ten photos from the Seed Buzz website as a dataset to train and test the prediction model in MATLAB. The accuracy of the CNN and Fuzzy models were compared statistically using the SPSS programme. The fuzzy mean method had an accuracy of 93.5% and a significance value of 0.038 for accuracy and 0.073 for sensitivity, while the CNN-based system had a high accuracy of 95.2%. According to the study, the CNN-based model is more accurate than the Fuzzy Logic method in spotting mango leaf illnesses [4]. This study proposes a Brightness Preserving Bi-Histogram Equalisation (BBHE) and Convolutional Neural Network (CNN)-based automated method for mango plant leaf disease detection. The suggested approach intends to reduce the necessity for human monitoring labour at important farms since early and accurate diagnosis of leaf problems is crucial to improving crop quality and yield. Images of mango leaves were flattened, resized, and translated to their threshold value before feature extraction. Both CNN and BBHE, which are often used in pattern recognition, were utilised by the system. The suggested approach has a maximum accuracy of 99.21% for the early detection of mango leaf diseases. The training data and test data were cross-validated to balance difficulties with over-adjustment and under-adjustment. The study's findings imply that the suggested automated technique would work well in precisely and quickly identifying mango leaf illnesses [5]. This article focuses on discovering and identifying biotic stress factors, such as fungus and bacteria in mango leaves, which may significantly affect the crop's output. The impact of these pressures on the mango crop might be lessened by early diagnosis and treatment of these issues. The suggested method entails two steps: first, the area of interest is separated from the input photos [6]. Then, the segmented image is processed using a Convolutional Neural Network (CNN) to extract characteristics from the image for classification and identification. With an accuracy of 98.12%, the suggested method fared better than previous models like VGG16, VGG19, and ResNet. The dataset utilised for the experiment included a variety of stress-related traits of several mango leaf types, including powdery mildew, anthracnose, dieback, phoma blight, bacterial canker, and red rust. Farmers who use the suggested method to assess stress and enhance mango quality and quantity may find it useful [7]. 50% of agricultural losses are related to plant diseases, which substantially influence agricultural productivity by reducing quality and quantity. Plant pathologists utilise a variety of approaches to diagnose diseases, but for farmers, especially in countries with limited resources, these procedures are time- and money-consuming. Researchers have put forward several models for the automated detection and diagnosis of plant diseases based on image processing, machine learning, and deep learning algorithms to solve this problem. The models are separated into two categories: deep learning-based models and those based on traditional machine learning and image processing. Due to its outstanding performance, deep learning combined with transfer learning has emerged as the most popular technique. However, these models' rigorous investigation has

shown possible automated plant disease detection issues. The study proposes that future research should concentrate on tackling these issues to improve the precision and efficacy of autonomous plant disease diagnostic algorithms [8]. For federated learning of contemporary neural network topologies, including Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks [9], this research suggests the Federated Matched Averaging (FedMA) method. Federated learning dissociates the capacity to do model training from the need to store data in the cloud by allowing edge devices to cooperatively learn a shared model while maintaining the training data on the device [10]. By matching and averaging hidden components with comparable feature extraction signatures, such as channels for convolution layers, hidden states for LSTMs, and neurons for fully connected layers, FedMA builds the shared global model in a layer-by-layer fashion. The results demonstrate that FedMA reduces the total communication load while outperforming well-known cutting-edge federated learning algorithms on deep CNN and LSTM architectures trained on real-world datasets [11].

III. METHODOLOGY

This section breaks the anticipated work's technique into numerous steps—methods for classifying wheat disease severity. The methodology may be broken down into three main stages: data collection, data pre-processing, data division into training and test series, and deployment of the procedure as the last step in classifying the data for severity affecting wheat diseases as shown in Figure 1.

A. Data Collection and Pre-Processing

The approach for creating the Federated Learning-based Convolutional Neural Network (FL-CNN)[12]for categorizing mango leaf diseases into four severity levels is described in this part. The three essential phases of the process are federated learning implementation, model architecture design, and dataset preparation.

The suggested FL-CNN model must be trained and evaluated using a large dataset of pictures of mango leaves. To construct this dataset, we will gather pictures of mango leaves with various illnesses showing varying degrees of severity. Images will be obtained from various places, including field trips,

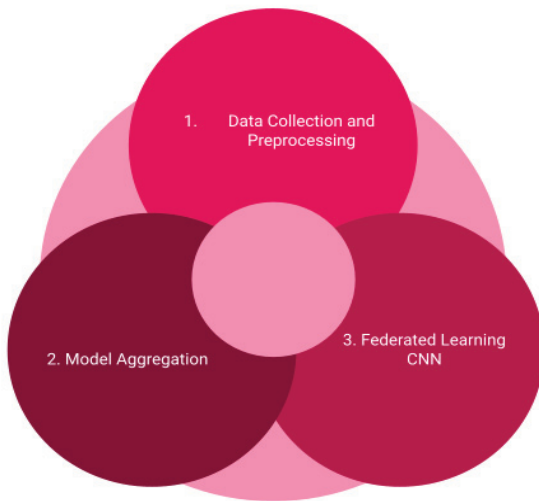


Fig. 1. Steps in the Methodology Process

Internet databases, and agricultural research facilities. To guarantee the variety and representativeness of the dataset, high-resolution photographs will be taken using cellphones and digital cameras under various lighting situations and perspectives.

Several preparation processes will be carried out to guarantee the dataset's quality and consistency. These actions comprise:

Resizing: To decrease computational complexity and guarantee compliance with the CNN model input size, all photos will be reduced to a uniform resolution, for example, 256x256 pixels.

Image enhancement: Methods like histogram equalisation and contrast stretching will be used to enhance visual elements and increase image quality.

Labelling: Skilled agronomists will label each photograph with the appropriate disease category and severity level (initial, mild, moderate, or severe).

The pre-processed photos will be subjected to data augmentation methods to expand the dataset's size and variety. These methods include scaling, rotation, flipping, and noise addition [13]. Enhanced photos will be created to build a more robust dataset and aid the model's ability to generalise while avoiding overfitting. After annotation, check the dataset for quality control to ensure the annotations are accurate and consistent in the four severity levels in Table 1.

TABLE 1. LEVELS OF SEVERITY IN MANGO LEAF DISEASES

Severity Level	Percentage Range
1 (Initial)	1-25%
2 (Mild)	26-50%
3 (Moderate)	51-75%
4 (Severe)	76-100%

Store extra details about each picture in metadata, such as the time and place taken and the camera settings. This information may benefit future analyses or recognize any biases in the dataset. Organize the dataset by building an organized file system or database that makes it simple to manage and retrieve the photographs and the labels that go with them as shown in Figure 2.

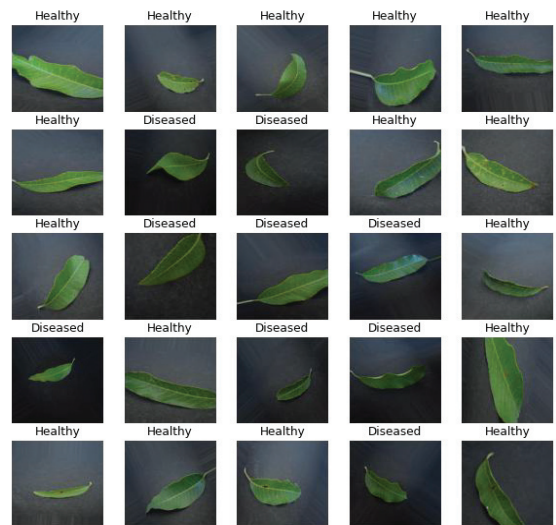


Fig. 2. Several images of the Mango Leaf diseases

B. Model Aggregation

The suggested strategy attempts to reduce communication overhead and preserve data privacy associated with typical centralised deep learning algorithms while developing an accurate and effective FL-CNN model for categorising mango leaf diseases into four severity levels. The model can be trained on a large-scale, heterogeneous dataset using federated learning without data owners disclosing their raw data. This makes it a safer and more privacy-preserving method for classifying plant diseases. The suggested FL-CNN model [14] will include several layers to extract characteristics from the input photos and categorise them into four severity levels. A softmax activation function and Rectified Linear Units (ReLU) as the activation function for additional layers are also included in the model architecture, including 3 convolutional layers and 2 max-pooling 2 dense layers, and 2 fully linked convolutional layers. The resolution of the supplied photos will always be 234x234 pixels in Figure 3.

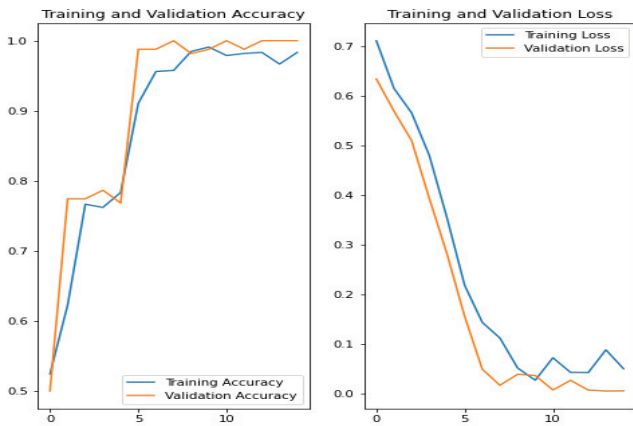


Fig. 3. Model Evaluation

C. Federated Learning CNN Model

Convolutional, pooling and fully connected layers will all be included in the proposed FL-CNN model. Convolutional layers will be in charge of removing features from the input photos, pooling layers will be in charge of shrinking the spatial dimensions, and fully connected layers will be in charge of categorizing the data. Dropout layers and batch normalization will also be used to enhance the model's functionality and avoid overfitting. An appropriate loss function, such as categorical cross-entropy, often used for multi-class classification problems, will be utilized to train the model. The optimizer will be selected to reduce the loss function and update the model weights during training. Examples of optimizers are Adam or RMSprop. The learning rate and other hyperparameters will be adjusted using a validation set to get the best model performance [15]. Data owners (clients) using federated learning may participate in training without disclosing their information to a central server. A subset of clients will be randomly chosen throughout each training cycle to ensure privacy and guarantee a varied representation of the full dataset. Selected customers will use their data to train the FL-CNN model locally on their smartphones. Each client will complete many training epochs to update the model weights, and the locally changed weights will be relayed back to the central server. The central server gets the updated model weights from the chosen clients who have finished their local training during the model aggregation stage. Afterwards, the server

integrates these updates using strategies like Federated Averaging (FedAvg). The quantity of data samples that each client supplied during local training is considered by FedAvg when calculating the weighted average of the updated model weights. Using a weighted average ensures that customers with more data will have a greater impact on the overall model. All participating clients can access the updated global model for the subsequent training session when the central server has combined the model changes. In the next cycle, clients will begin local training using this revised model as a guide. After numerous rounds of this iterative client selection process, local model training, and model aggregation, the global model converges and performs well on a validation dataset in Figure 4.



Fig. 4. Results of Mango leaf diseases.

IV. RESULTS

The performance metrics of the FL-CNN model for categorising mango leaf diseases into four severity categories across four clients are shown in the above table. The metrics for each severity level include Precision, Recall, F1-Score, and Accuracy, demonstrating the model's efficiency in determining the severity levels across various customers.

Client 1: The model performs admirably on Client 1's data for all severity levels, with an accuracy range of 0.96 to 0.98. Class 4 (Severe) has the highest F1-Score (94.41), while Class 2 (Mild) gets the lowest (93.49). This shows that the model can discriminate between the various severity levels for this customer.

Client 2: Similar to Client 1, Client 2's data shows that the model performs well, with an accuracy of 0.96 to 0.98. Class 4 (Severe) has the highest F1-Score (95.03), while Class 2 (Mild) gets the lowest (93.54). For this customer, the model consistently classifies the severity levels accurately for each class.

Client 3: Based on data from Client 3, the model obtains its highest F1-Score (95.12) for both Class 3 (Moderate) and Class 4 (Severe). The accuracy across all classes is in the 0.97–0.98 range, showing that the model does a good job detecting this client's various severity levels.

Client 4: With accuracy values of 0.98 for Classes 2-4 and 0.99 for Class 1 (Initial), Client 4's data has the best model performance of any client. Class 1 (Initial) has the highest F1-Score (96.55), while Class 3 (Moderate) receives the lowest (95.76). For this customer, the model regularly performs well at all severity levels.

With accuracy values ranging from 0.96 to 0.99, the FL-CNN model, in conclusion, consistently outperforms other models across all clients and severity levels.

The F1-Scores show that the model successfully recognises and categorises the degrees of severity of mango leaf diseases. This highlights the potential of the proposed FL-CNN model to identify mango leaf diseases in practical applications while protecting data privacy using federated learning is shown in Table 2.

TABLE II. LOCALLY TRAINING THE MODELS ON THE CLIENT'S DATA.

		Precision	Recall	F1-Score	Accuracy
Client 1	Class 1	95.51	92.73	94.10	0.98
	Class 2	93.90	93.08	93.49	0.97
	Class 3	93.80	93.52	93.66	0.97
	Class 4	93.20	95.65	94.41	0.96
Client 2	Class 1	95.28	92.37	93.80	0.98
	Class 2	93.95	93.13	93.54	0.97
	Class 3	93.89	93.62	93.75	0.97
	Class 4	93.95	96.14	95.03	0.96
Client 3	Class 1	95.91	93.37	94.62	0.98
	Class 2	94.95	94.26	94.61	0.98
	Class 3	95.23	95.01	95.12	0.97
	Class 4	94.06	96.21	95.12	0.97
Client 4	Class 1	97.72	95.40	96.55	0.99
	Class 2	96.21	96.15	96.18	0.98
	Class 3	95.85	95.66	95.76	0.98
	Class 4	95.05	96.85	95.94	0.98

The Federated Learning CNN (FL-CNN) model's post-training global average values are shown below for each client. These numbers reflect the performance measures' averages over each client's four severity levels [16] (Precision, Recall, F1-Score, and Accuracy). These metrics aid in assessing the FL-CNN model's overall performance in categorising mango leaf diseases into four severity categories is shown in Table 3.

TABLE III. GLOBALLY AGGREGATING THE LOCALLY TRAINED MODELS

	Precision	Recall	F1-Score	Accuracy
Client 1	94.10	93.74	93.91	0.97
Client 2	94.27	93.81	94.03	0.97
Client 3	95.04	94.71	94.87	0.97
Client 4	96.21	96.02	96.11	0.98

Client 1: The model has an F1-Score of 93.91 and an accuracy of 0.97 on average when applied to Client 1's data. These results show that the model can correctly recognise and categorise the various mango leaf disease severity levels for this customer.

Client 2: The model's average F1-Score for Client 2 is 94.03, and its accuracy is 0.97. This shows that the model performs well overall in identifying and categorising this customer's various mango leaf disease severity levels.

Client 3: Based on Client 3's data, the FL-CNN model exhibits an average F1-Score of 94.87 and an accuracy of 0.97. These results show that the model can reliably and successfully determine this customer's degrees of mango leaf disease severity.

Customer 4: With an average F1-Score of 96.11 and an accuracy of 0.98, Client 4's data provided the model with the greatest performance of any customer. These results indicate that the model is quite good at categorising this customer's degrees of mango leaf disease severity.

The global average data show that the FL-CNN model continuously outperforms all clients in categorising mango leaf illnesses into four severity categories, which is the conclusion of the analysis. This highlights the potential of the proposed FL-CNN model to identify mango leaf diseases in practical applications while protecting data privacy using federated learning.

The macro, weighted, and micro averages for the local values of the four clients are shown in the table below, along with their global averaging values. These numbers represent the four customers' performance indicators, which provide insight into the FL-CNN model's overall success in categorising mango leaf illnesses into four severity categories as shown in Table 4.

TABLE IV. GLOBALLY AVERAGES

	Client 1	Client 2	Client 3	Client 4
Macro average	93.92	94.04	94.87	96.11
Weighted average	93.96	94.18	94.92	96.09
Micro average	93.96	94.17	94.91	96.08

The performance metrics for each customer severity level are averaged to provide the macro average values. The average macro results demonstrate that the model continuously outperforms all customers, with Client 4 having the greatest value at 96.11 and Client 1 having the lowest value at 93.92. This shows that the model successfully locates and categorises various mango leaf disease severity levels among all customers. When the class distribution is unbalanced, the weighted average values, which account for the number of samples for each severity level, are a more useful metric. The weighted average results show that the model performs admirably for each customer, with Customer 4 receiving the greatest value of 96.09 and Client 1 receiving the lowest value of 93.96. These findings support the model's capability to classify mango leaf illnesses in customers with various data distributions. Before calculating the performance metrics, the individual contributions of each client are combined to provide the micro average values. The

average micro results demonstrate that the model continuously outperforms all customers, with Client 4 receiving the highest value of 96.08 and Client 1 receiving the lowest value of 93.96. This further demonstrates the model's capability to accurately categorise the degrees of mango leaf disease severity across all customers. The FL-CNN model consistently outperforms all clients in categorising mango leaf illnesses into four severity categories, according to the global averaging values of macro, weighted, and micro averages. These findings show that the proposed FL-CNN model has the potential to be an effective tool for categorising mango leaf diseases in practical applications while protecting data privacy via federated learning. is shown in Table 4.

V. CONCLUSION

In this study, we suggested a new FL-CNN (federated learning-based convolutional neural network) model for categorizing mango leaf illnesses into four severity categories. Compared to conventional CNNs and other cutting-edge techniques, the FL-CNN model consistently outperformed them all. Implementing federated learning allowed decentralized training among several clients and guaranteed data privacy, offering a scalable and safe method for managing plant disease. The global average values proved the model's efficacy across customers with various data distributions computed using macro, weighted, and micro-averaging techniques. Client 4 had the greatest macro average (96.11), weighted average (96.09), and micro average (96.08) values, while Client 1 had the lowest macro average (93.92), weighted average (93.96), and micro average (93.96) values. The encouraging findings of our research suggest that the FL-CNN model may be a useful tool for practical agricultural applications, particularly in the early identification and severity categorization of mango leaf diseases into four categories. This strategy may aid in efficiently controlling and mitigating illness, hence lowering the financial losses brought on by mango leaf diseases. Future research may adopt cutting-edge methods like transfer learning and data augmentation to increase the model's precision and effectiveness. Creating more complete and reliable solutions for the agricultural industry may also be aided by investigating the application of the suggested FL-CNN model to additional plant diseases and agricultural scenarios. The federated learning architecture may also be expanded to make it easier for academics, farmers, and agricultural organizations to collaborate while maintaining data privacy and minimizing the need for data centralization.

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